Bilinear and Quadratic Forms

Problem sheet for the Day 10 class · CSIR NET previous year questions and exam-pattern problems

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. CSIR NET Dec 2015 · MCQ

Consider the quadratic form $Q(v) = v^t A v$, where $A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$, $v = (x, y, z, w)$. Then

(A) Q has rank 3

(B) $xy + z^2 = Q(Pv)$ for some invertible 4×4 real matrix P

(C) $xy + y^2 + z^2 = Q(Pv)$ for some invertible 4×4 real matrix P

(D) $x^2 + y^2 - zw = Q(Pv)$ for some invertible 4×4 real matrix P

2. Exam pattern · MCQ

For the real quadratic form $Q(x, y, z) = 2xy + 2yz + 2zx$, the numbers of positive and negative eigenvalues of its matrix are

(A) 1 and 2 (B) 2 and 1

(C) 3 and 0 (D) 1 and 1

3. Exam pattern · MCQ

The quadratic form $x^2 + y^2 + z^2 + 2k(xy + yz + zx)$ on $\mathbb{R}^3$ is positive definite if and only if

(A) $-\frac{1}{2} < k < 1$ (B) $0 < k < 1$

(C) $|k| < 1$ (D) $k > -\frac{1}{2}$

4. Exam pattern · MCQ

The number of congruence classes of real symmetric 3×3 matrices (equivalently, of real quadratic forms in 3 variables up to an invertible change of variables) is

(A) 10 (B) 9

(C) 4 (D) 7

5. Exam pattern · NAT

The rank of the real quadratic form $Q(x_1, x_2, x_3, x_4) = x_1x_2 + x_2x_3 + x_3x_4$ is _____.

6. CSIR NET June 2018 MSQ Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 1 \end{pmatrix}$ and define for $x, y, z \in \mathbb{R}$, $Q(x, y, z) = (x\ y\ z)A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Which of the following statements are true?

(A) The matrix of second order partial derivatives of the quadratic form Q is $2A$

(B) The rank of the quadratic form Q is 2

(C) The signature of the quadratic form Q is $(+\ +\ 0)$

(D) The quadratic form Q takes the value 0 for some non-zero vector (x, y, z)

7. CSIR NET Dec 2014 MSQ Let f be a non-zero symmetric bilinear form on $\mathbb{R}^3$. Suppose that there exist linear transformations $T_i : \mathbb{R}^3 \to \mathbb{R}, i = 1, 2$ such that for all $\alpha, \beta \in \mathbb{R}^3$, $f(\alpha, \beta) = T_1(\alpha)T_2(\beta)$. Then

(A) $\mathbf{rank}\ f = 1$

(B) $\dim\{\beta \in \mathbb{R}^3 : f(\alpha, \beta) = 0 \text{ for all } \alpha \in \mathbb{R}^3\} = 2$

(C) f is a positive semi-definite or negative semi-definite

(D) $\{\alpha : f(\alpha, \alpha) = 0\}$ is a linear subspace of dimension 2

8. CSIR NET June 2019 MSQ Consider the matrix $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ over the field of rationals. Which of the following matrices are of the form P^tAP for a suitable 2×2 invertible matrix P over $\mathbb{Q}$? Here P^t denotes the transpose of P.

(A) $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$

(B) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(D) $\begin{pmatrix} 3 & 4 \\ 4 & 5 \end{pmatrix}$

9. CSIR NET June 2019 MSQ Consider a matrix $A = (a_{ij})_{5\times5}$, $1 \le i, j \le 5$ such that $a_{ij} = \frac{1}{n_i + n_j + 1}$, where $n_i, n_j \in \mathbb{N}$. Then in which of the following cases A is a positive definite matrix?

(A) $n_i = i$ for all $i = 1, 2, 3, 4, 5$

(B) $n_1 < n_2 < \cdots < n_5$

(C) $n_1 = n_2 = \cdots = n_5$

(D) $n_1 > n_2 > \cdots > n_5$

10. Exam pattern MSQ For $A \in M_n(\mathbb{R})$ let $B(u, v) = u^tAv$ be the associated bilinear form on $\mathbb{R}^n$. Which of the following are true?

(A) B is symmetric if and only if A is symmetric

(B) $B(u, u) = 0$ for all u if and only if A is skew-symmetric

(C) Every bilinear form on $\mathbb{R}^n$ is uniquely the sum of a symmetric and an alternating bilinear form

(D) If B is non-degenerate, then $B(u, u) \neq 0$ for some u